DEEP LEARNING

Chest X-ray Denoising

A classical filter, a convolutional autoencoder and a U-Net, compared on NIH ChestX-ray14

Andrea Fascì · A.y. 2026-2027



Why denoise chest X-rays?

A lower radiation dose is safer for the patient, but fewer photons reach the detector and the image becomes grainy.

Do convolutional networks clearly beat a classical filter, and do skip connections preserve fine details?

H1

Both networks beat a Gaussian filter tuned on validation data.

H2

Same network + skip connections (U-Net) beats the plain autoencoder.



Reference image



Simulated low dose

Data: NIH ChestX-ray14

4,999
X-rays, first of the 12 archives

→ keep frontal (PA) views

3,290
images of 1,264 patients

→ preprocess

128×128
grayscale, values in [0, 1]

Split by patient, 70 / 15 / 15

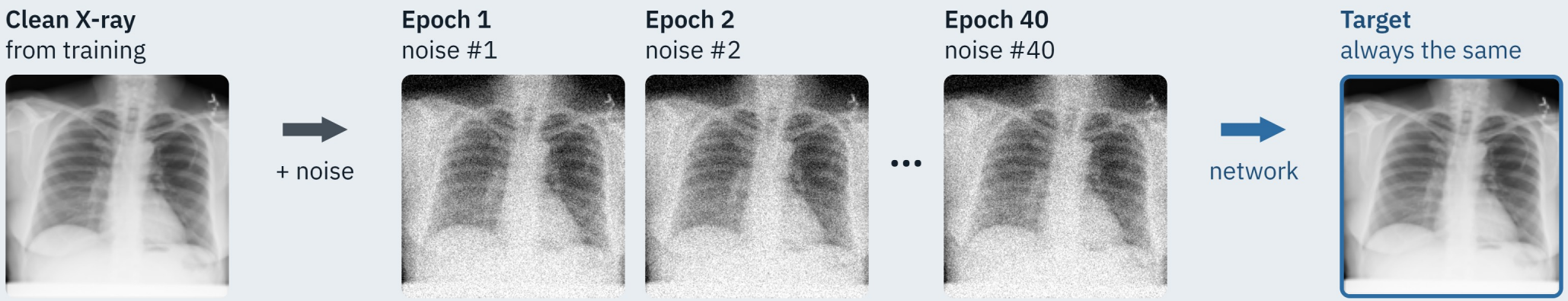
Training 2,274	Validation 503	Test 513
the networks learn	best epoch and filter σ	final scores, used once

Why by patient?

Many patients have several X-rays. If one went to training and another to test, the network would be tested on a chest it has already seen.

The noise never repeats

Additive Gaussian noise, σ = 0.1, clipped to [0, 1]: an approximation of low-dose noise.



~91,000 noisy inputs from 2,274 X-rays: the noise never repeats, the anatomy does.

Training: new noise at every step

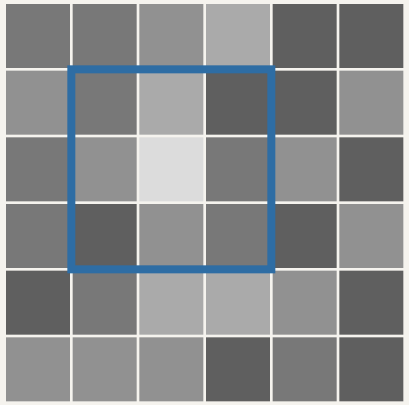
The network cannot memorize the noise, only learn the anatomy: free data augmentation.

Validation and test: fixed noise

Generated once with fixed seeds: every method is judged on the same noisy images.

Baseline: the Gaussian filter

X-ray pixels



Pixel + neighbours

5	7	4
6	**9**	5
4	6	5

×

Weights (÷ 16)

1	**2**	**1**
2	**4**	**2**
1	**2**	**1**

=

Its new value

6.4

1. Put a window on one pixel and its neighbours.

2. Weighted average: closer neighbours count more. That is the pixel's new value.

3. Repeat for every pixel: a new, filtered image of the same size.

Corners 18 × 1 + sides 24 × 2 + centre 9 × 4 = 102, and 102 ÷ 16 = 6.4: the noisy 9 drops.

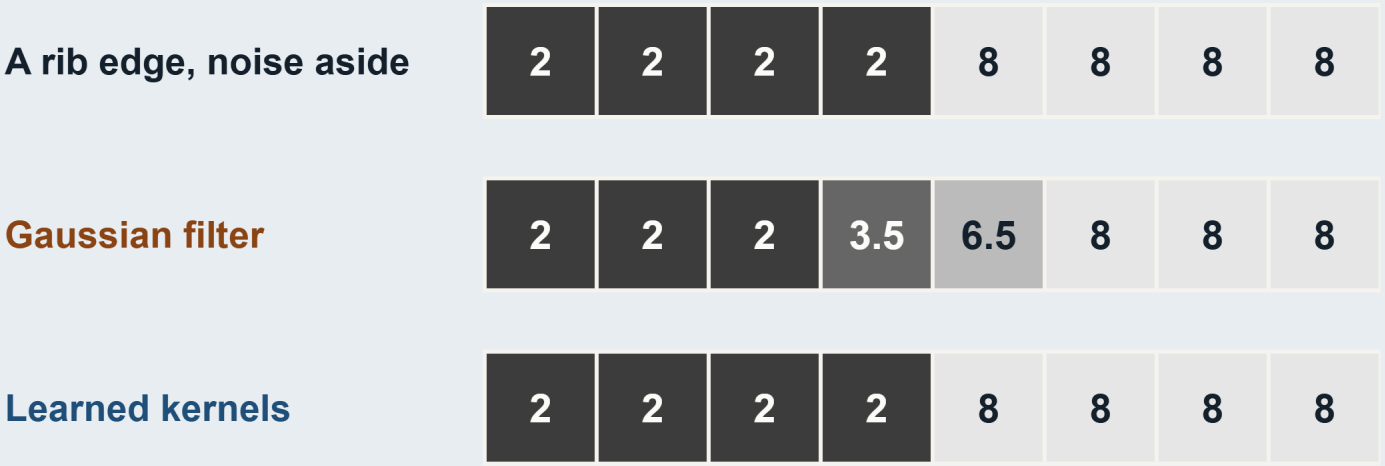
In our project

An 11×11 window (σ = 1.25 px), on all 16,384 pixels of each X-ray. Applied once: no training.

Compared fairly with the networks

σ chosen on validation among 6 values. Then the same 513 noisy test images and the same MSE.

Fixed kernel or learned kernels?



One row of pixels across a rib edge: dark lung, then bright bone.

Filter: near the edge the window covers both sides, so the sharp step becomes a ramp.

Learned kernels: some detect edges, others flat areas, so the step can stay sharp.

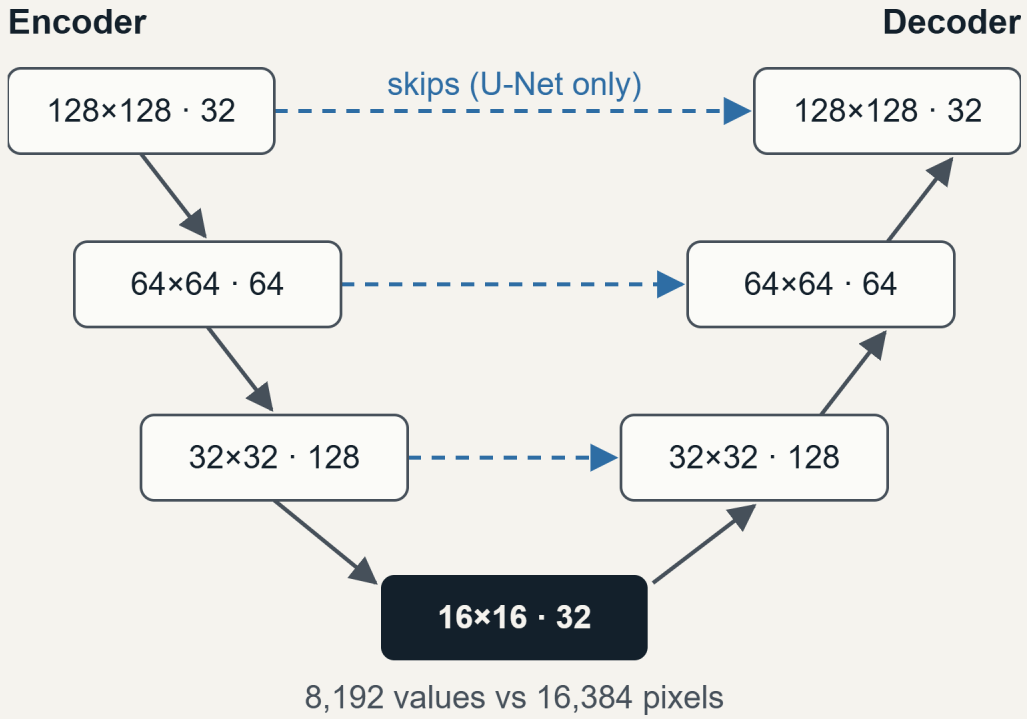
Gaussian filter: one kernel, fixed

Its weights come from a formula, the same everywhere: noise and edges get the same average.

Networks: hundreds of kernels, learned

Same operation, a convolution. Their weights start random and backpropagation sets them.

The two networks: one shape, one difference



1. Encoder and decoder

Each level: two 3×3 convolutions, then max-pool. The size halves and the filters double, 32 → 64 → 128. The decoder mirrors it back to 128×128.

2. Bottleneck

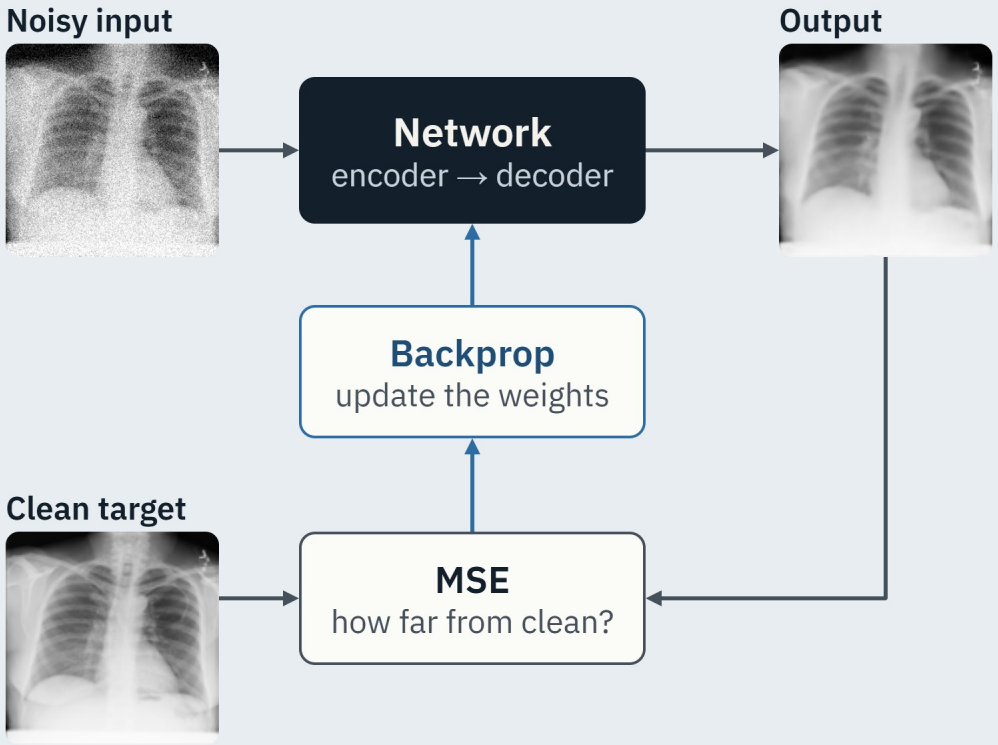
16×16×32 = 8,192 values, fewer than the 16,384 pixels: the autoencoder cannot copy, it must compress.

3. Skip connections, U-Net only

They carry fine details past the bottleneck. The only difference between the two networks.

Convolutions exploit locality, stationarity and compositionality: **0.74 M weights** for the whole autoencoder, vs 268 M for one dense 128×128 layer.

Training: the same for both networks



Repeated with new noise at every step.

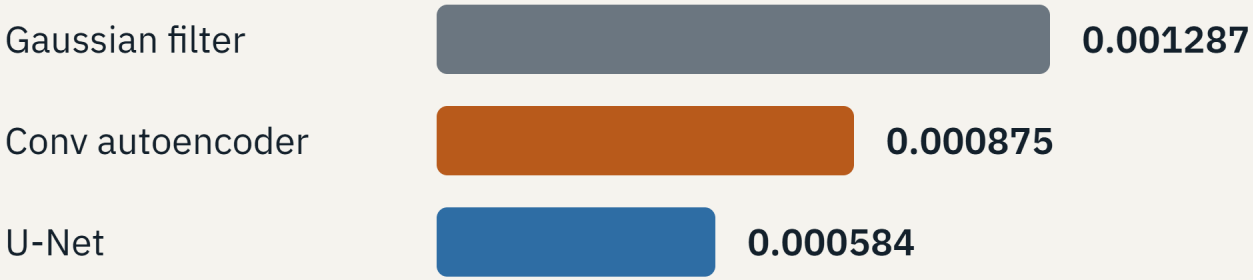
To lower the MSE the network must predict the clean X-ray. Anatomy repeats across X-rays, noise does not: so it keeps the anatomy and drops the noise.

Loss	MSE, a pixel-wise regression
Optimizer	Adam, learning rate 10^{-3}
Batch	32 X-rays
Epochs	40, best validation epoch kept
Seeds	0, 1, 2, so the result does not depend on the start

Standard settings, not tuned, identical for both: only the skip connections differ.

Results on 513 test images

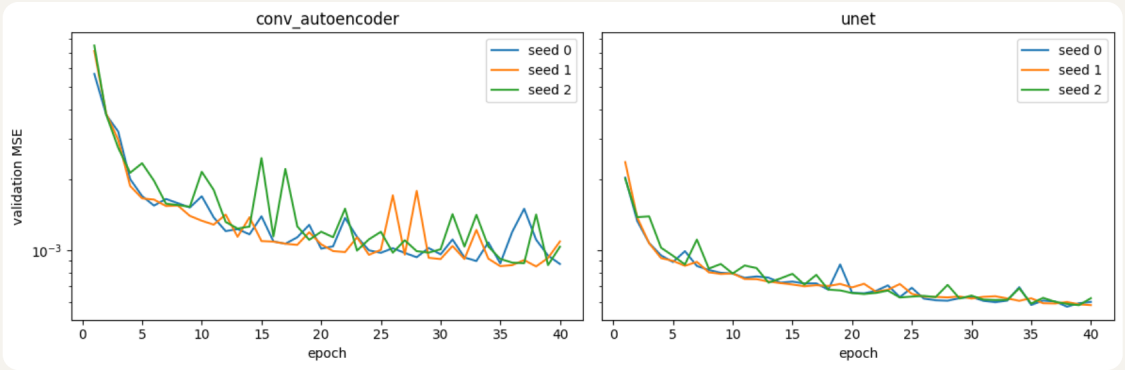
Test MSE (lower is better)



Starting point, noisy input: 0.0092 (off the scale)

U-Net vs filter: error 2.2 times smaller (H1)

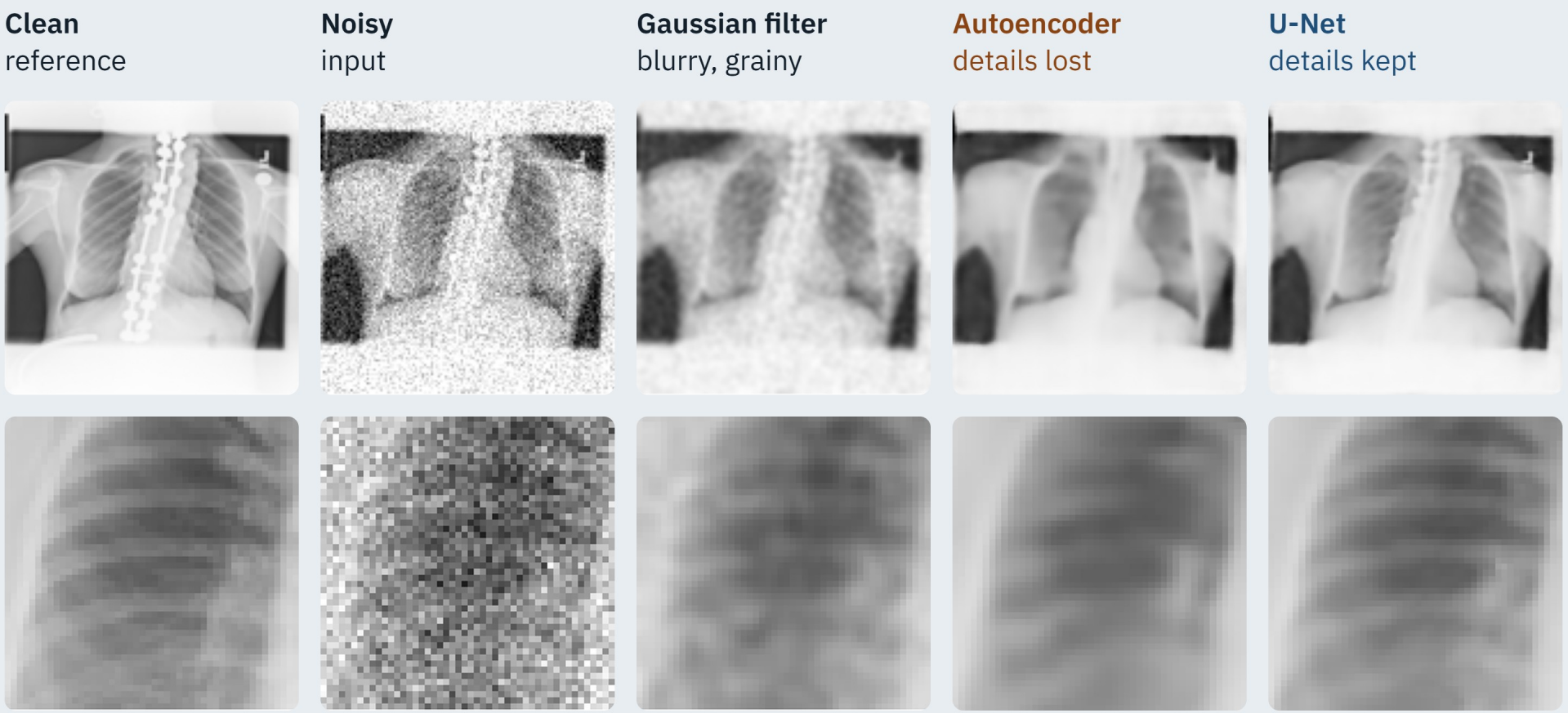
U-Net vs autoencoder: error 33% smaller (H2)



Validation curves, 3 seeds each: the U-Net learns faster and more stably.

Same data, layers, training budget and noisy test images for all methods. Seed-to-seed std ≤ 0.00001.

Visual evidence



Chest X-ray denoising · 10 · top: test X-ray with a spinal implant · bottom: 48 × 48 zoom on the ribs

Conclusions

H1 confirmed

The U-Net error is 2.2× lower than the tuned Gaussian filter's: learning from data removes noise without blurring the edges.

H2 confirmed

Same layers plus skip connections: 33% less error and sharper details, a gap about 40× the seed-to-seed spread.

Limitations

- One random Gaussian noise, the same in training and test: structured noise or a different test noise not tested
- The U-Net has 26% more parameters
- Best epochs 38–40: longer training could help
- Low MSE ≠ small lesions preserved

Thank you. Questions?